What Sets the Readable Depth of a Placed Ring, and How Far a Self-Checking Layer Pushes It: Scan, Register, Avoid and Repair on Three Heavy-Hex Devices Without the Calibration Map

TN Holdings
love.os.architect@proton.me
https://github.com/TN-Holdings-LLC/psf-zero

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Abstract

A circuit placed on a superconducting device stays readable only to some depth. In earlier work we measured that depth without reading the calibration map and found, on `ibm_kingston`, a stretch of a ring whose echo was near random or inverted while the calibration showed nothing there. Here we ask what sets the readable depth on a given day and device, and how much of it a user-side layer can recover from its own runs alone. The layer tests the rings it would use with a mirror echo read per site, keeps a list of the paths it found dead (listed after two jobs, expired after seven days), places the next ring around them with the same formula, and, only for mirror-type circuits, applies a local frame flip that leaves the ideal circuit unchanged. All tests were declared before their fetch, and the placement is fold:rule against fold:rule throughout. Two effects are kept apart. *Width*: where a dead run sat on the standard ring, avoiding it made every site usable again (kingston $n = 40$: 22 of 40 to 40 of 40 on three days; $n = 72$: 38–54 to 72; WIDER in all six declared comparisons, and the same 18-qubit run on every day), and on two more devices the same scan-then-avoid widened the ring (`ibm_fez` $n = 40$: 29 to 40, $n = 72$: 36 to 63; `ibm_marrakesh` $n = 72$: 67 to 72). *Accuracy*: on a ring that was already fully usable, a forward Clifford computation with a known answer was slightly more right on the avoiding ring (marrakesh, $n = 72$: mean site signal 0.776 to 0.816 at 12 layers, 0.643 to 0.694 at 24), and all of the gain sat in the ten qubits that the avoiding ring replaced. The ring avoiding the registry reached the depth of the $n = 20$ control at $n = 40$ but not at $n = 72$, where a shortfall of about two layers remained with no weak run visible to the scan. A two-qubit frame flip lifted kingston's dead block by +0.32 and +0.30 on two days without cost elsewhere, but recovered it only to about a third of the healthy sites at 24 layers. Avoiding comes first; repair is for a ring that cannot avoid the path.

1 Introduction

- **The question.** Paper 5 [1] measured how deep a ring stays readable on IBM's heavy-hex devices when no placement reads a quality number, as the decay length L^*, and reported a defect that no calibration number shows. Here we ask two things: (i) what sets that depth on a given day and device, and (ii) how much of it a layer on the user's side can recover using only its own runs.
- **An analogy, for the reader only.** An operating system tests its memory (memtest), keeps a list of bad blocks, allocates around them, and sometimes patches around a fault. The layer

here does the same with a mirror echo, a list of dead paths, the ring placement and a frame flip. Like Paper 5's acoustic picture, this is an analogy, not an identity.

- **Two effects, kept apart from the start.** *Width*: how many ring sites still carry information (usable sites). *Accuracy*: how right the output is on a ring that is already fully usable. The same "$+x$" can mean either; every result in Section 5 says which one it is.

- **What is new.** (1) A self-measured defect registry that holds paths, not qubits, with dates and expiry, built from the user's own jobs and no quality number. (2) Declared tests of scan, then avoid, on three devices. (3) The separation of width and accuracy gains, including a forward computation. (4) Repair by a local frame flip, with its limits.

- **What is not new.** Mirror circuits as benchmarks [5]; layer fidelity and errors per layered gate [6]; noise-aware layout from calibration data (for example Qiskit's VF2 post-layout pass, which scores layouts with the target's error rates [7]); fluctuations of qubit quality in time from two-level defects [8]; Pauli-frame and twirling identities [9]. The placement formula and its comparison with other placements are Paper 4 [2] and Paper 5 [1]; they are not repeated here.

2 Yardsticks

Five quantities appear below. They answer different questions, and their numbers are never put side by side as "deeper" or "shallower". Ratios are taken only within one row (standard against avoiding ring, same device and day).

	what is read	depth unit	records	typical rule values
Y1 charge L^*	four-input ratio of the weight-2 charge	one-way steps L	29 Oct	kingston 81–90, marrakesh 56–57
Y2 echo L^*	four-input ratio ρ_E of the mirror echo	echo layers D (both ways)	30 Oct on	kingston 22–26 ($n = 20$, 36)
Y3 site signal	per-site echo signal a_j at $D = 24$	—	30 Oct on	healthy 0.35–0.45
Y4 forward signal	per-site signal of an identity Clifford circuit	layers D	1 Nov	marrakesh $n = 72$: 0.78–0.82 ($D = 12$)
Y5 parity V^*	weight-n parity, fitted on nL	nL	30 Oct	kingston 73–106

Table 1: The yardsticks. Y1 and Y2 differ both in the observable (a conserved weight-2 charge against a local echo that spreads) and in the unit (one-way steps against both-way layers).

- **Site classes (Y3).** A site is *dead* if $a_j < 0.12$; a dead run is at least three consecutive dead ring sites (cyclic). A site is *weak* if $a_j < \max(0.12,\ 0.5 \times \text{ring median})$; a weak run likewise. The *usable* sites are those outside every dead run. These rules were written before the coverage-days declaration (TOQB `toqb/defect_list.py`, `toqb/safe_loop.py`) and are used unchanged.

- **Registry entry.** The four qubits at the centre of a dead run (a *path*). Two sightings sharing two qubits are the same defect. A path is LISTED after two jobs and EXPIRED seven days after it was last seen.

3 What sets the readable depth

3.1 The placement

Held fixed here (the rule). Its advantage over the other map-free placements, in ΔL and in 0 losses of 144 echo comparisons, is Paper 5's result [1].

3.2 The observable

- The charge is conserved by the circuit and stays a weight-2 quantity; its L^* (Y1) is long.
- The echo's local Z_j spreads within the light cone of the circuit; its L^* (Y2) is shorter, but at $D = 24$ the rule's ρ_E did not shrink with n up to 72 (0.17–0.44 across devices and sizes).
- The parity has weight n; at $n \geq 40$ its readout reference itself fell towards zero, so the parity's size wall on these devices was set by readout.
- **A failed prediction.** A trajectory twin with T_1, T_2 and static ZZ predicted the echo at 0.08–0.10 for $n = 20$, $D = 24$, and below the noise floor at $n = 40$ and 72. The device gave 0.17–0.44.

3.3 Defects on the ring

- **Black.** The same 18 qubits (3–7, 17, 27–29, 38, 49–55, 59) formed a dead run on the standard $n = 40$ ring of `ibm_kingston` in every run of the byte-identical circuit: on 30 October, 31 October, 1 November and 2 November (Figure 2). Seventeen of these 18 qubits were healthy on other rings the same day, so the defect belongs to a path, not to the qubits.
- **Gray.** A weak stretch (35, 19, 15, 14) on an $n = 72$ ring read 0.10–0.12 at 11:40 CET and 0.46–0.49 at 13:14 CET on 31 October, with the same circuit on the same qubits.
- **A coherent part.** Dynamical decoupling did not lift the dead sites, and gate twirling turned their inversion into noise; the block's signal crossed zero over depth (between $D = 16$ and $D = 20$); a frame flip brought part of it back (Section 5.4).
- **Chip, and partly position.** The byte-identical $n = 20$ circuit had no dead run on kingston and lost 11 of 20 sites on `ibm_fez`. On the other hand, some ring positions (positions 57–64 of the $n = 72$ ring) were weak on all three chips. The defects depend on the chip, and some positions repeat across chips.

3.4 A size effect with no visible defect

On rings with no weak run at all, the rings of $n = 48$–72 read about two layers shallower than the $n = 20$ control of the same job (Y2 L^* 21.3–22.3 against 24.1), and the $n = 72$ ring placed around the registry was SHORTER by the clean-depth test (20.4–20.8 against 23.9–24.7). We do not know why; it is reported as open.

4 The self-checking layer

- **Map-free.** No calibration quality number is read; only the coupling graph, the basis and the failed elements.

layer	function	records
L1 scan	mirror echo per site on the rule's standard rings ($n = 20, 40, 72$; $D = 24$)	coverage days, two devices
L2 registry	black paths (LISTED after 2 jobs, 7-day expiry); gray paths for the day only	`list_after.json` files
L3 allocate	the rule's placement with the middle coupler of each listed path marked failed	coverage B, clean depth, M3
L4 repair	local frame flip, mirror-type circuits only, entries verified on two days	black stages 2–3, fix, fix repeat

Table 2: The layers used in the declared tests. A sentinel layer (echo circuits riding in the user's job) is planned and not yet tested.

- **Self-measured.** Every registry entry comes from the user's own runs, with dates.
- **Avoid by default.** Repair only with an entry verified on two days, and only for mirror circuits.
- **Black carried, gray not.** Black entries carry over days; gray ones apply on the day they were seen.
- **No cause chased.** An entry is kept or dropped by what it does.
- **Declared.** Each claim is a declared test (Table 7), with stop rules that weigh the size of effects (Section 5.5).
- **Cost.** A scan is 24 circuits; it was billed 9 s per device, and is skipped when the registry already holds a scan of the day.

5 Results

5.1 Width: usable sites (Y3)

device, day	$n = 40$: standard → avoiding	$n = 72$: standard → avoiding	verdicts
kingston, 31 Oct (coverage day 1)	22 → 40	38 → 72	WIDER, WIDER
kingston, 1 Nov (coverage day 2)	22 → 40	39 → 72	WIDER, WIDER
fez, 1 Nov	29 → 40	36 → 63	HELPS
marrakesh, 1 Nov	40 → 40	67 → 72	HELPS
kingston, 2 Nov (coverage day 3)	22 → 40	54 → 72	WIDER, WIDER

Table 3: Usable sites on the standard ring and on the ring placed around the registry, same device and day.

- **The show-off number and the essential one.** The show-off number is usable sites 22 → 40. The essential one: on the sites that were already usable, the mean signal moved only 0.41 → 0.44 (day 1), 0.39 → 0.44 (day 2) and 0.41 → 0.43 (day 3). The gain is width. The whole-ring mean (0.17 → 0.44) mixes both effects and is not used as a headline.
- **The defect stayed where it was.** On all three days the listed path [28, 29, 38, 49] was dead on the standard ring (PRESENT, 3 of 3: PERSISTENT by the declaration), and the avoiding rings had no dead run. Of the two $n = 72$ paths, [11, 10, 9] was present on all three days, while [87, 86, 85] was present on days 1 and 2 and absent on day 3; it stayed listed (seven-day expiry) and was avoided with nothing to avoid.

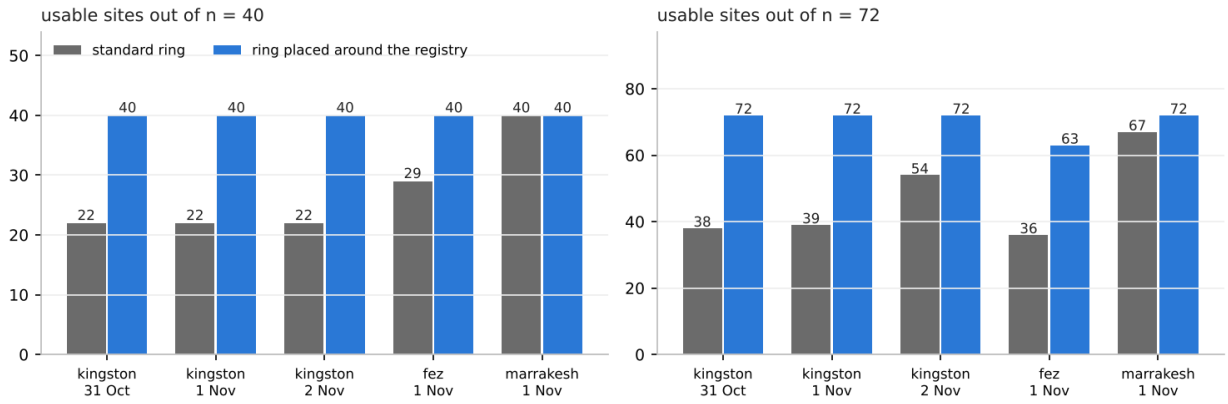


Figure 1: Usable sites, standard ring (grey) against the ring placed around the registry (blue). From the coverage files of each test.

- **Not everything was masked.** On fez the avoiding $n = 72$ ring still carried a nine-site dead run next to a listed path; it entered the registry for the next day.
- **When there is nothing to avoid.** On marrakesh at $n = 40$ the standard ring had no dead run, and the avoiding ring changed nothing (SAME).
- **Three days, by the declaration.** C WIDER in 6 of 6 and NARROWER in none; P PERSISTENT. The sentences fixed in advance: placing around a self-measured list widens the usable part of the device; the defect stays for days without showing in the calibration, so a list with a multi-day expiry is sound.

5.2 Depth recovered (Y2, rings placed around the registry)

ring	standard ring (30 Oct)	avoiding ring (31 Oct)	verdict against the $n = 20$ control
kingston $n = 40$	13.0–16.0	24.2–24.6	REACHED (3 of 3 h_z)
kingston $n = 72$	14.6–16.0	20.4–20.8	SHORTER (3 of 3 h_z)

Table 4: Echo L^* (layers D, Y2) on kingston. The $n = 20$ control read 23.9–24.7 in the same job.

- **A stricter list does not close the gap at $n = 72$.** With weak runs listed too (the safe ladder), the safe size of that day was $n = 40$; $n = 48$ to 72 had no weak run and were still about two layers short, so the scan cannot see what remains (Section 3).

5.3 Accuracy: a forward computation on a fully usable ring (Y4)

- **Circuit.** D layers alternating (S on all, CZ ring) and (H on all, CZ ring). It is the identity whenever D is a multiple of 6, at every ring length, and it is not a mirror: no layer is undone by a later one. The answer is the input, bit by bit, on four inputs.
- **Test.** `ibm_marrakesh`, $n = 72$, the standard ring (holding the listed coupler (27, 28)) against the ring placed around the registry. Both jobs ran within one second of each other.

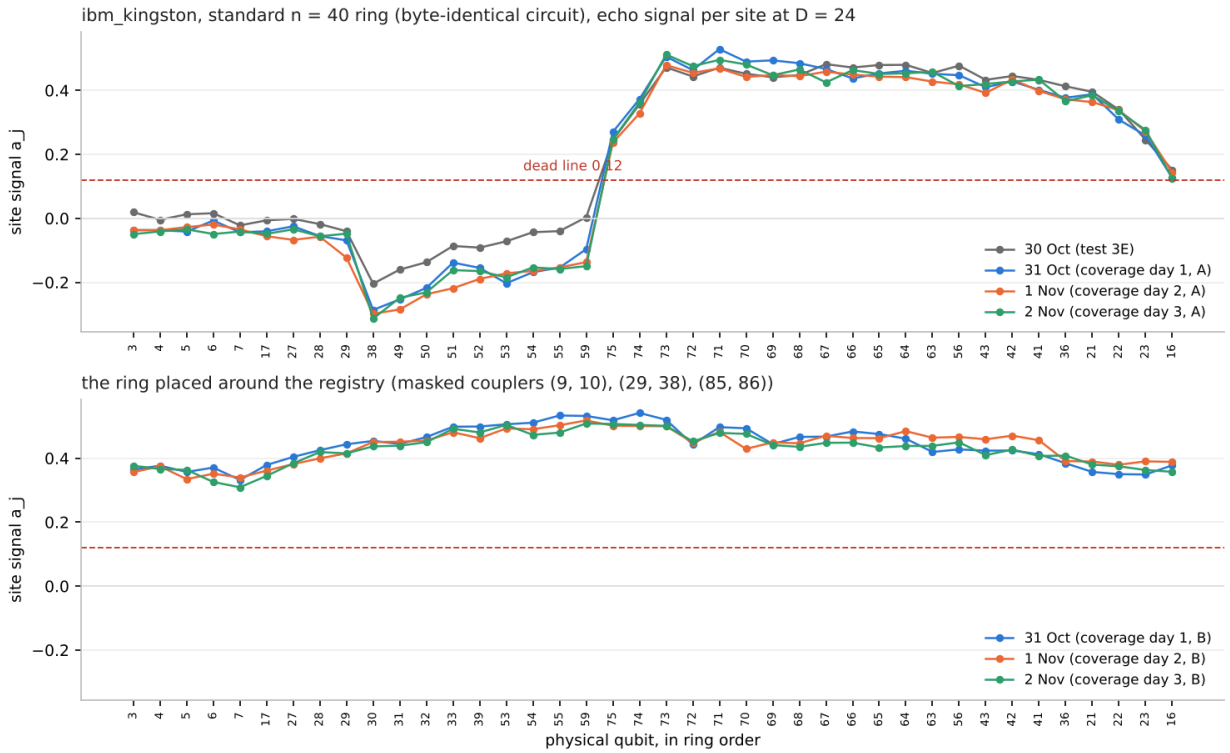


Figure 2: Top: the per-site echo signal along kingston's standard $n = 40$ ring on four days (byte-identical circuit). The run 3–59 is below the dead line every time. Bottom: the ring placed around the registry, same days.

D	standard	avoiding	difference	usable sites
12 (judged)	0.776	0.816	$+0.040 \pm 0.002$	$72 \to 72$
24 (reported)	0.643	0.694	$+0.052 \pm 0.002$	$72 \to 72$

Table 5: Mean site signal of the forward computation. Verdicts at $D = 12$: Q1 HELPS, Q2 SAME.

- **The show-off number and the essential one.** The show-off number is per-bit right 88.8 % $\to$ 90.8 %. The essential ones: the gain is 5 % and 8 % of the signal, and all of it sits in the ten swapped qubits (0.615 $\to$ 0.866 at $D = 12$). On the 62 shared qubits the two rings read the same (0.802 against 0.808).
- **What kind of gain.** The listed path was weak that day, not dead, so nothing was unusable on the standard ring. The gain is accuracy, not width.
- **The first attempt (kingston) did not reach its question.** It used n layers of (H, CZ ring), the identity only after $D = n$ layers. At $n = 40$ this meant 40 layers, beyond the readable depth of either ring: 0.009 on the standard ring and 0.020 on the avoiding one, under the declared floor of 0.05, so the result is INCONCLUSIVE. The $n = 20$ control read 0.268. This was a design error. The lesson: a forward test needs a period shorter than L^*.

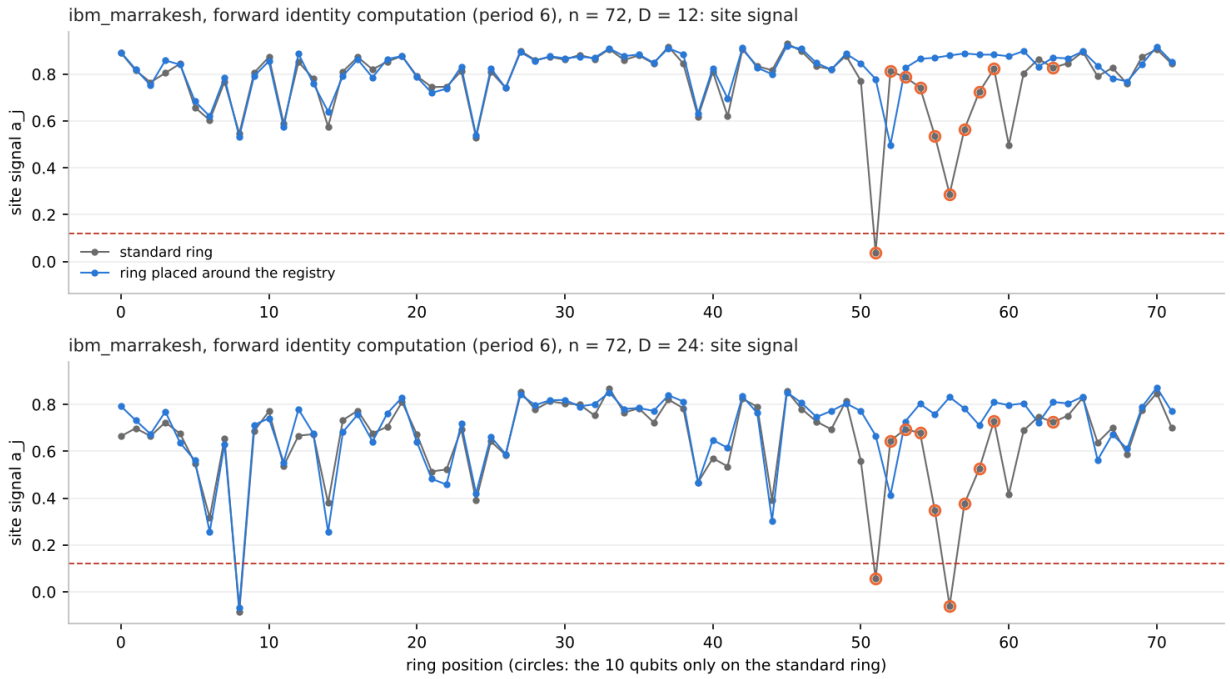


Figure 3: Site signal of the forward computation, standard ring (grey) and avoiding ring (blue). Circles: the ten qubits only on the standard ring.

5.4 Repair: a local frame flip

- **The flip.** X on chosen ring positions just before and after the CZ ring of every second layer, with a virtual Z on the neighbours that need it. The ideal circuit is unchanged as an operator.
- **Global against local.** A flip on all even positions lifted kingston's dead block but cost the healthy part of the ring. A flip around the block alone did nothing. Bisection then traced the gain to the first part of the run, and a flip on qubits 27 and 29 alone carried it.

day	block, off $\to$ flip at 27, 29 ($D = 24$)	difference	rest of the ring
31 Oct	$-0.193 \to +0.129$	$+0.322 \pm 0.007$	$+0.008$ (NO COST)
1 Nov	$-0.185 \to +0.112$	$+0.298 \pm 0.007$	$+0.005$ (NO COST)

Table 6: The two-qubit flip on kingston's dead block (38, 49–55, 59). PERSISTS on the second day.

- **Limits.** The block was not RECOVERED (0.8 $\times$ the rest). It reached about 0.30 $\times$ at $D = 24$ and 0.75 $\times$ at $D = 12$. In usable sites the flip gave $22 \to 25$ at $D = 24$ and $32 \to 36$ at $D = 12$, against 40 of 40 on the avoiding ring.
- **Mirror circuits only.** In simulation the flip cancels a static ZZ error exactly only in a mirror circuit; for a forward computation it gives nothing, so the repair was not tried on one.
- **Conclusion.** Avoid first; repair only when the ring cannot avoid the path.

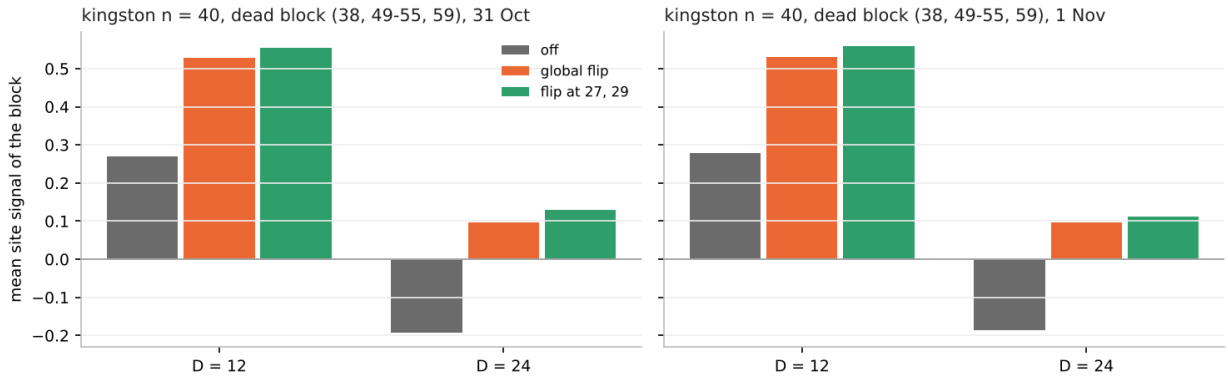


Figure 4: Mean signal of kingston's dead block, flip off, global flip, and flip at 27 and 29, on two days.

5.5 Procedure failures, reported as results

- **Bisection stop rule.** The rule stopped on "both halves HELP", with +0.019 beside +0.299. Stop rules have weighed the size of effects since then.
- **Gray, stage 1.** Its precondition, the weak stretch, was absent when the test ran (Section 3), so nothing was judged.
- **Forward test on kingston.** A design error (Section 5.3).

6 Discussion

- **Width against accuracy.** Avoiding pays in width when a dead run lies on the standard ring (kingston $n = 40$, 72; fez). It pays a little in accuracy when the path is only weak (marrakesh $n = 72$). It changes nothing when the standard ring is clean (marrakesh $n = 40$).
- **What set the readable depth on these days.** With the placement held fixed: defects on the ring (on kingston $n = 40$, Y2 L^* 13–16 on the standard ring against 24 around the registry), the weight of what is read, a residual size effect of about two layers at $n \geq 48$, and gray drift within hours. The order of the last three is not established.
- **Why map-free matters here.** The calibration showed nothing at kingston's dead run [1]. A layout scored by calibration error rates would not have avoided it.
- **Psf-os.** The layers are being written as an open-source layer (AGPL-3.0) between a ring circuit and the device. Its skeleton reproduces the registry of the records byte for byte, and places the same rings as the test code. No performance of it is claimed here.

7 Limitations

- Three IBM heavy-hex devices, three days at most; one circuit family (the rule's ring); one forward computation, on one device and one day.
- The registry's thresholds are fixed choices, not fitted; other choices would list other paths.

- No error correction, no pulse access, and no cause of a defect is claimed.
- At $n = 72$ the avoiding ring did not reach the $n = 20$ depth; we do not know why.
- Repair works for mirror circuits only, and is weak at $D = 24$.
- Standard errors are shot noise only.

8 Reproducibility

TOQB [4]: `results/device-2026-10-30`, `results/device-2026-10-31-fez`, `results/coverage-days-2026-10-31-day1`, `results/coverage-days-2026-11-01-day2`, `results/clean-depth-2026-10-31`, `results/safe-ladder-2026-10-31`, `results/gray-stage1-2026-10-31`, `results/black-stage1-2026-10-31` to `results/black-fix-2026-10-31`, `results/fix-repeat-2026-11-01`, `results/two-devices-2026-11-01`, `results/qca-m3-2026-11-01`, `results/qca-m3b-2026-11-01`, `results/defect-list-replay-2026-10-31`; declarations in `prereg/`. The figures and every derived number are made by `readable_depth_self_check_figures.py` from those records.

test	declaration	SHA-256 (start)	fixed (CET)
coverage days	`coverage_days_note.md`	4941ddfa	31 Oct 11:29
clean depth	`clean_depth_note.md`	2c45ee4b	31 Oct 11:55
safe ladder	`safe_ladder_note.md`	67d10447	31 Oct 12:27
gray stage 1	`gray_stage1_note.md`	25f6231a	31 Oct 13:04
black stage 1	`black_stage1_note.md`	04737044	31 Oct 13:22
black stage 2	`black_stage2_note.md`	72aedffa	31 Oct 13:41
black stage 3	`black_stage3_note.md`	11f59455	31 Oct 13:58
black bisection	`black_bisect_note.md`	3697ce48	31 Oct 15:09
black fix	`black_fix_note.md`	0c72bbe8	31 Oct 15:30
fix repeat	`fix_repeat_note.md`	935d3a5d	1 Nov 03:04
two devices	`two_devices_note.md`	91f2dbb8	1 Nov 11:30
M3 (kingston)	`qca_m3_note.md`	8a5618ca	1 Nov 12:35
M3, second form	`qca_m3b_note.md`	e3337f85	1 Nov 15:48
coverage day 3	(the coverage-days note)	4941ddfa	31 Oct 11:29

Table 7: Declarations, each fixed before the test's fetch.

A Calibration-only placement: a lower bound and a construction

This appendix summarises a TOQB note [4] (`notes/separation_2026-11-02/`, not declared). Its numbers are simulations on a fake device, not device results.

Lower bound. Let $C(\theta)$ be the calibration numbers of a device state θ: static per-element numbers and the outcomes of *isolated* experiments, in which no two couplers at distance one carry a two-qubit gate in the same layer. A policy that reads only the coupling graph and C gives the same placement law to all states with the same C. Hence, for states $\theta_1, \dots, \theta_M$ with equal C, its worst expected regret is at least $\frac{1}{M} \min_L \sum_k R_{\theta_k}(L)$ (the averaged two-point method of Le Cam [10]). This is close to a tautology. If a coupler e carries an error only when a coupler at distance one is driven in the same layer, and a ring through e costs Δ, the bound over all e gives at least $n\Delta/|E_w|$ for a ring of n qubits ($|E_w| = 164$ working couplers on FakeKingston: 0.12Δ, 0.24Δ, 0.44Δ at

$n = 20, 40, 72$). The bound does not cover policies that read same-layer measurements (e.g. layer fidelity [6]), the results of the user's own circuits, or their own tests of the day.

Upper bound, with its assumptions. psf-os's scan reads each site from 4×1000 outcomes in $\{-1, +1\}$; by Hoeffding's inequality [11] a site at distance t from the line 0.12 is misread with probability at most $2e^{-2Nt^2}$ (3×10^{-13} one-sided at $t = 0.12$, $N = 1000$). This bounds the regret only if (A1) a dead echo means a bad user circuit, (A2) the cost depends only on whether the ring holds a dead path, (A3) the comparator is the rule's own placement with every dead path masked, not the best placement, and (A4) nothing changes between scan and job. Near the line the bound says nothing: at $a_j = 0.10$ it is 0.90. The line 0.12, the three sites and the two jobs are chosen values.

Construction (Aer, FakeKingston, one planted coupler, no other noise).

check	planted state	baseline
the coupler alone ($m = 2, 8, 32$)	counts identical to the baseline	1.000
with a neighbour in the same layer	0.746, 0.269, 0.250 (Pauli)	1.000
Pauli ZZ 0.15, scan at the Clifford point	usable 24 $\to$ 40 ($n = 40$), 56 $\to$ 72 ($n = 72$)	40, 72
coherent $R_{ZZ}(\pi/8)$, $n = 20$, $h = 0.6$	usable 20 $\to$ 20, ring mean 0.40 $\to$ 1.00	1.000
local frame flip (coherent)	every site 1.000	1.000

The error is invisible to the isolated benchmark and visible at once to a same-layer one, so it separates the two kinds of policy. Two limits are part of the result. In the coherent model the planted stretch read 0.10, just under the line; it was read as dead in three of four readings of the same circuit and missed in one, so detection there is close to chance. And the model's local flip repairs the stretch completely with no cost, while on `ibm_kingston` the flip at 27, 29 repaired the block only in part (Section 5); the device defect is consistent with this construction but is not explained by a single static ZZ term.

Acknowledgments

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